# Tharwat Farag Badreldin

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## Professional Summary

Microsoft Certified Data Engineer (DP-700, DP-600, DP-800, DP-900) with 4+ years of hands-on experience in SQL and relational databases, including database administration and production troubleshooting. Completed the Digilians Data Engineering Program at the Egyptian Military Academy in collaboration with the Ministry of Communications and Information Technology (MCIT), developing practical experience in ETL/ELT, data pipelines, data modeling, Python, SQL, Microsoft Fabric, and Azure. Previously led the company's E-learning/EdTech sector at Camp Coding, overseeing 100+ educational platforms serving more than 130,000 active production users.

## Education

### B.Sc. Computer Science — Sep 2019 – Jun 2023

*Faculty of Computers and Information, Tanta University* — *Tanta, Egypt*

- Very Good Grade.

## Work Experience

### Data Engineering Specialist Trainee — Jan 2026 – Jun 2026

*Digilians · Egyptian Military Academy (in collaboration with MCIT)* — *Cairo, Egypt*

- Built on 4+ years of prior hands-on SQL and database experience to develop and optimize ETL/ELT pipelines using Python and SQL.
- Improved SQL query performance through query optimization and indexing strategies.
- Designed scalable relational data models for analytical systems.
- Performed data extraction, transformation, and loading (ETL) to support business intelligence and reporting.
- Collaborated with cross-functional teams to deliver reliable and scalable data solutions.

### Technical Lead – LMS & EdTech Platforms — Jul 2020 – Dec 2025

*Camp Coding* — *Tanta, Egypt*

- Owned the full lifecycle of the company's E-learning/EdTech sector, managing client relationships from requirements gathering and commercial negotiation through platform delivery and ongoing technical support.
- Led the development and deployment of 100+ educational/LMS platforms for clients across Egypt and the Middle East, supporting more than 130,000 active users in production environments.
- Investigated and resolved production data issues directly at the SQL/database level, performing data validation, record consistency checks, database troubleshooting, root-cause analysis, and hands-on database administration for live platforms.
- Troubleshot backend-level issues by investigating application files and configuration when problems could not be resolved through the application interface or database.
- Led and coordinated the technical team, assigning responsibilities, following up on tasks, and resolving internal and client-facing technical escalations.
- Maintained direct post-launch relationships with clients and students, handling requirements, technical issues, data-related problems, feedback, and ongoing platform support.

### IT Manager — Nov 2023 – Jul 2025

*Baseeta* — *Part-time*

- Oversaw IT operations, improving system reliability and operational efficiency.
- Led website optimization initiatives focused on performance and security.
- Collaborated with stakeholders to align technology solutions with business objectives.

### Software Sales Specialist — Dec 2024 – Jun 2025

*Alnaierh* — *Riyadh, Saudi Arabia · Remote*

- Maintained software operations, ensuring high system availability and stability.
- Contributed to website lifecycle management and performance enhancement.
- Bridged communication between technical teams and business stakeholders.

## Key Projects

### B.Laban Retail Customer & Sales Analytics Platform — *GitHub*

- Built an end-to-end retail data engineering platform using PySpark, Microsoft Fabric, Delta Lake, and SQL with a Bronze → Silver → Gold medallion architecture.
- Implemented data-quality validation, anomaly flagging, standardization, and automated testing across 7K transactional records, identifying 48 invalid transactions while preserving source-data traceability.
- Designed business-ready Gold datasets for branch, product, channel, customer, and daily-sales analytics, with prepared outputs for downstream reporting.

### SSIS Data Warehouse ETL Pipeline — *GitHub*

- Built an end-to-end Data Warehouse solution using SSIS, SQL Server, and Star Schema architecture.
- Implemented ETL workflows with SCD Type 1 & 2 handling and integrated SSAS/SSRS reporting.

### Real Estate Data Platform — *GitHub*

- Built an end-to-end real estate data engineering pipeline using Python, web scraping, Pandas, SQL Server, and analytical SQL views.
- Implemented data cleaning and quality validation for sale and rental listings, producing business KPIs and dashboard-ready datasets.

### Airflow ETL Pipeline — *GitHub*

- Developed automated ETL workflows using Apache Airflow and Astronomer in a Dockerized environment.

### Hospital Admissions Pipeline — *GitHub*

- Built an end-to-end Big Data pipeline transforming raw hospital data into analytics datasets using Python, Hive, and Docker.

## Technical Skills

**Programming & Query Languages:** Python, SQL, JavaScript

**Database Platforms:** SQL Server, PostgreSQL, MySQL, SQLite

**SQL & Database Engineering:** SQL Development, Database Administration, Database Troubleshooting, Data Validation, Data Integrity, Query Optimization, Indexing, Database Deployment, Database Security

**Data Engineering:** ETL/ELT Pipelines, Data Pipelines, Data Extraction, Data Transformation, Data Loading, Data Warehousing, Data Modeling, Relational Data Models, Star Schema, Apache Airflow, SSIS, Docker

**Microsoft Fabric:** Data Factory, Data Pipelines, Lakehouse, OneLake, Data Warehouse, Notebooks, Semantic Models

**Big Data & Cloud:** Apache Spark, Apache Kafka, Hadoop, Hive, Microsoft Azure

**Data Analytics & BI:** Business Intelligence (BI), SSAS, SSRS, Analytical SQL

**Data Processing & Tools:** Pandas, NumPy, Selenium, Git, GitHub, Jupyter Notebook, Visual Studio Code

**Leadership:** Technical Leadership, Team Leadership, Client Management, Stakeholder Management, Requirements Gathering, Cross-functional Collaboration, System Architecture

## Certifications

Microsoft Certified: Fabric Data Engineer Associate (DP-700)

Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)

Microsoft Certified: SQL AI Developer Associate (DP-800)

Microsoft Certified: Azure Data Fundamentals (DP-900)

## Languages

**Arabic:** Native

**English:** Professional Working Proficiency